

Pharmaceutical Claims Database

Entity-Relationship Design Document

Health Insurance Analytics Platform

Version 1.0 | Designed for Post-Adjudication Analytics

1. Design Overview

This document describes the logical entity-relationship design for a pharmaceutical claims analytics database supporting a health insurance organization. The database is architected for post-adjudication analysis, with primary use cases including drug cost decomposition (plan paid vs. member cost share vs. other payer), medication adherence measurement using Proportion of Days Covered (PDC) at the NDC level with therapeutic grouping, formulary compliance analysis, and member-level accumulator tracking.

Key design decisions incorporated into this schema:

- Member dimension uses Slowly Changing Dimension Type 2 (SCD2) to preserve full history of demographic and eligibility changes across time.
- Claim records support full adjustment and reversal history; analytics must filter to the latest adjudicated version per claim.
- Formulary is versioned with effective/expiration dates at both the drug-tier assignment level and the tier cost sharing rules level.
- NDC-to-GPI-14 drug reference table is maintained internally with package and unit information.
- PDC therapeutic groupings map NDCs to cohorts, enabling cross-NDC adherence measurement within a drug class.
- Accumulators track individual and family deductible/out-of-pocket progress with cross-accumulator contribution linkage.
- Prior authorization and formulary exception lifecycle is fully tracked with approval/denial status, effective dates, and approving clinician.

2. Entity Inventory

The following entities comprise the database schema, organized by domain:

Entity Name	Domain	Purpose
MEMBER_DIM	Member	Versioned member demographic and eligibility record (SCD2)
MEMBER_ENROLLMENT	Member	Plan enrollment periods per member version
EMPLOYER	Member	Employer group information
HEALTH_PLAN	Plan	Health plan definition and metadata
FORMULARY	Plan	Versioned formulary definition
FORMULARY_DRUG	Plan	Drug-to-tier assignments within a formulary version
FORMULARY_TIER	Plan	Tier cost sharing rules within a formulary version
ACCUMULATOR	Member	Member benefit period accumulator buckets (individual/family)
ACCUMULATOR_CONTRIBUTION	Member	Cross-accumulator contribution linkage
CLAIM	Claims	Core pharmaceutical claim record
CLAIM_COST	Claims	Cost decomposition per claim (plan, member, other payer)
CLAIM_ADJUSTMENT	Claims	Adjustment and reversal history per claim
DRUG_REFERENCE	Drug	NDC master with GPI-14 mapping, package, and unit info
PDC_THERAPEUTIC_GROUP	Drug	Therapeutic cohort groupings for PDC calculation
PDC_NDC_GROUP_MAP	Drug	NDC-to-therapeutic-group mapping
PRESCRIBER	Provider	Prescriber NPI, name, specialty, demographics
PHARMACY	Provider	Pharmacy NPI, name, type, location
PRIOR_AUTH	Authorization	Prior authorization and formulary exception lifecycle

3. Entity Definitions

3.1 Member Domain

MEMBER_DIM

The core member entity implementing SCD Type 2. Each change to a member's demographic or eligibility attributes creates a new row. The MEMBER_ID remains constant across all versions; MEMBER_SK is the system-generated surrogate key uniquely identifying each version. Claims reference MEMBER_SK to point to the exact version active at fill date.

MEMBER_DIM			
Column Name	Data Type	Constraint	Description
Primary Key			
MEMBER_SK	BIGINT	PK, NOT NULL	System-generated surrogate key — unique per version
Attributes			
MEMBER_ID	VARCHAR(20)	NOT NULL, INDEX	Constant natural member identifier across all versions
FIRST_NAME	VARCHAR(50)	NOT NULL	Member first name
LAST_NAME	VARCHAR(50)	NOT NULL	Member last name
DATE_OF_BIRTH	DATE	NOT NULL	Member date of birth
GENDER	CHAR(1)	NOT NULL	Member gender code
MARITAL_STATUS	VARCHAR(20)		Marital status at this version
ADDRESS_LINE1	VARCHAR(100)		Street address
CITY	VARCHAR(50)		City
STATE	CHAR(2)		State code
ZIP_CODE	VARCHAR(10)		ZIP / postal code
EMPLOYER_ID	INT	FK → EMPLOYER	Employer associated with this version
EFFECTIVE_DATE	DATE	NOT NULL	Date this version became active
EXPIRATION_DATE	DATE		Date this version was superseded (NULL = current)
IS_CURRENT	BOOLEAN	NOT NULL	Flag: TRUE if this is the current active version
CREATED_AT	TIMESTAMP	NOT NULL	Row creation timestamp

Notes: SCD2: when a member attribute changes, IS_CURRENT on the prior row is set to FALSE and EXPIRATION_DATE is set to the day before the new version's EFFECTIVE_DATE.

MEMBER_ENROLLMENT

Tracks a member's plan enrollment periods. A member may have multiple enrollment periods over time, each tied to a specific health plan and coverage type. This entity is separate from MEMBER_DIM so that plan changes do not necessarily trigger a new member version.

MEMBER_ENROLLMENT			
Column Name	Data Type	Constraint	Description
Primary Key			
ENROLLMENT_ID	BIGINT	PK, NOT NULL	System-generated enrollment period identifier
Foreign Keys			
MEMBER_SK	BIGINT	FK → MEMBER_DIM	Member version active at enrollment start
MEMBER_ID	VARCHAR(20)	NOT NULL, INDEX	Constant member ID for cross-version joins
HEALTH_PLAN_ID	INT	FK → HEALTH_PLAN	Health plan for this enrollment period
Attributes			
COVERAGE_TYPE	VARCHAR(20)	NOT NULL	Individual, Employee+Spouse, Family, etc.
BENEFIT_PERIOD_START	DATE	NOT NULL	Start of the benefit / plan year
BENEFIT_PERIOD_END	DATE	NOT NULL	End of the benefit / plan year
ENROLLMENT_START_DATE	DATE	NOT NULL	Date member enrolled in this plan
ENROLLMENT_END_DATE	DATE		Date enrollment ended (NULL = currently enrolled)
GROUP_NUMBER	VARCHAR(20)		Employer group number
SUBSCRIBER_ID	VARCHAR(20)		Subscriber ID on the health plan

EMPLOYER

EMPLOYER			
Column Name	Data Type	Constraint	Description
Primary Key			
EMPLOYER_ID	INT	PK, NOT NULL	System-generated employer identifier
Attributes			
EMPLOYER_NAME	VARCHAR(100)	NOT NULL	Employer legal name
EMPLOYER_CODE	VARCHAR(20)		Internal or external employer code
INDUSTRY	VARCHAR(50)		Industry classification
ADDRESS_LINE1	VARCHAR(100)		Employer street address
CITY	VARCHAR(50)		City
STATE	CHAR(2)		State code
ZIP_CODE	VARCHAR(10)		ZIP code
CREATED_AT	TIMESTAMP	NOT NULL	Row creation timestamp

3.2 Plan Domain

HEALTH_PLAN

HEALTH_PLAN			
Column Name	Data Type	Constraint	Description
Primary Key			
HEALTH_PLAN_ID	INT	PK, NOT NULL	System-generated health plan identifier
Attributes			
PLAN_NAME	VARCHAR(100)	NOT NULL	Full plan name
PLAN_CODE	VARCHAR(20)	NOT NULL, UNIQUE	Short plan code
PLAN_TYPE	VARCHAR(20)	NOT NULL	HMO, PPO, HDHP, EPO, POS, etc.
INSURANCE_COMPANY	VARCHAR(100)		Underwriting insurance company
DEDUCTIBLE_INDIVIDUAL	DECIMAL(10,2)		Individual deductible amount
DEDUCTIBLE_FAMILY	DECIMAL(10,2)		Family deductible amount
OOP_MAX_INDIVIDUAL	DECIMAL(10,2)		Individual out-of-pocket maximum
OOP_MAX_FAMILY	DECIMAL(10,2)		Family out-of-pocket maximum
ACTIVE_FLAG	BOOLEAN	NOT NULL	Whether this plan is currently offered
CREATED_AT	TIMESTAMP	NOT NULL	Row creation timestamp

FORMULARY

The versioned formulary header. Each formulary version has its own effective/expiration dates, enabling point-in-time lookups at claim fill date.

FORMULARY			
Column Name	Data Type	Constraint	Description
Primary Key			
FORMULARY_ID	INT	PK, NOT NULL	System-generated formulary version identifier
Foreign Keys			
HEALTH_PLAN_ID	INT	FK → HEALTH_PLAN	Plan this formulary belongs to
Attributes			
FORMULARY_NAME	VARCHAR(100)	NOT NULL	Descriptive formulary name
FORMULARY_VERSION	VARCHAR(20)		Version label (e.g., 2024-v2)
EFFECTIVE_DATE	DATE	NOT NULL	Date this formulary version became effective

EXPIRATION_DATE	DATE		Date this version was superseded (NULL = current)
IS_CURRENT	BOOLEAN	NOT NULL	TRUE if this is the current active version
CREATED_AT	TIMESTAMP	NOT NULL	Row creation timestamp

Notes: To find the formulary in effect at a claim fill date: join on HEALTH_PLAN_ID and where EFFECTIVE_DATE <= fill_date AND (EXPIRATION_DATE IS NULL OR EXPIRATION_DATE >= fill_date).

FORMULARY_DRUG

Maps individual NDCs (or GPIs) to formulary tiers within a specific formulary version. Versioned via the parent FORMULARY record.

FORMULARY_DRUG			
Column Name	Data Type	Constraint	Description
Primary Key			
FORMULARY_DRUG_ID	BIGINT	PK, NOT NULL	System-generated identifier
Foreign Keys			
FORMULARY_ID	INT	FK → FORMULARY	Formulary version this assignment belongs to
NDC	CHAR(11)	FK → DRUG_REFERENCE	NDC being assigned to a tier
TIER_ID	INT	FK → FORMULARY_TIER	Tier this NDC is assigned to
Attributes			
COVERAGE_STATUS	VARCHAR(20)	NOT NULL	Covered, Non-Covered, Prior-Auth-Required, Step-Therapy
QUANTITY_LIMIT	DECIMAL(10,3)		Quantity limit if applicable
QUANTITY_LIMIT_DAYS	INT		Day supply period for quantity limit
STEP_THERAPY_REQUIRED	BOOLEAN		Whether step therapy applies
PRIOR_AUTH_REQUIRED	BOOLEAN		Whether prior authorization is required
EFFECTIVE_DATE	DATE	NOT NULL	Assignment effective date within formulary version
EXPIRATION_DATE	DATE		Assignment expiration date (NULL = active)

FORMULARY_TIER

Defines cost sharing rules per tier within a formulary version. Versioned via the parent FORMULARY record.

FORMULARY_TIER			
Column Name	Data Type	Constraint	Description

Primary Key			
TIER_ID	INT	PK, NOT NULL	System-generated tier identifier
Foreign Keys			
FORMULARY_ID	INT	FK → FORMULARY	Formulary version this tier belongs to
Attributes			
TIER_NUMBER	INT	NOT NULL	Tier number (1 = preferred generic, etc.)
TIER_LABEL	VARCHAR(50)		Descriptive label (e.g., Preferred Generic)
COPAY_RETAIL_30	DECIMAL(10,2)		Retail 30-day supply copay
COPAY_RETAIL_90	DECIMAL(10,2)		Retail 90-day supply copay
COPAY_MAIL_90	DECIMAL(10,2)		Mail order 90-day supply copay
COINSURANCE_PCT	DECIMAL(5,2)		Coinsurance percentage (0–100) if applicable
COINSURANCE_MIN	DECIMAL(10,2)		Minimum coinsurance dollar amount
COINSURANCE_MAX	DECIMAL(10,2)		Maximum coinsurance dollar amount
SUBJECT_TO_DEDUCTIBLE	BOOLEAN	NOT NULL	Whether cost sharing applies before deductible is met
EFFECTIVE_DATE	DATE	NOT NULL	Tier rule effective date
EXPIRATION_DATE	DATE		Tier rule expiration date (NULL = active)

3.3 Accumulator Domain

ACCUMULATOR

Tracks deductible and out-of-pocket accumulator progress per member per benefit period. Separate rows for individual and family accumulators. Each accumulator bucket has its own limit and accumulated amount.

ACCUMULATOR			
Column Name	Data Type	Constraint	Description
Primary Key			
ACCUMULATOR_ID	BIGINT	PK, NOT NULL	System-generated accumulator bucket identifier
Foreign Keys			
MEMBER_ID	VARCHAR(20)	NOT NULL, FK → MEMBER_DIM	Constant member ID (spans versions)
ENROLLMENT_ID	BIGINT	FK → MEMBER_ENROLLMENT	Enrollment period this accumulator belongs to
Attributes			
ACCUMULATOR_TYPE	VARCHAR(20)	NOT NULL	DEDUCTIBLE or OUT_OF_POCKET
ACCUMULATOR_LEVEL	VARCHAR(10)	NOT NULL	INDIVIDUAL or FAMILY
BENEFIT_PERIOD_START	DATE	NOT NULL	Benefit period start date
BENEFIT_PERIOD_END	DATE	NOT NULL	Benefit period end date
LIMIT_AMOUNT	DECIMAL(10,2)	NOT NULL	Maximum accumulator threshold
ACCUMULATED_AMOUNT	DECIMAL(10,2)	NOT NULL	Amount accumulated to date
AS_OF_DATE	DATE	NOT NULL	Date the accumulated amount was last updated

Notes: One row per accumulator type (DEDUCTIBLE / OOP) per level (INDIVIDUAL / FAMILY) per benefit period per member.

ACCUMULATOR_CONTRIBUTION

Records how a claim's member cost share contributes to accumulator buckets, supporting cross-accumulator relationships where individual spend also counts toward family limits.

ACCUMULATOR_CONTRIBUTION			
Column Name	Data Type	Constraint	Description
Primary Key			
CONTRIBUTION_ID	BIGINT	PK, NOT NULL	System-generated contribution identifier
Foreign Keys			
CLAIM_ID	BIGINT	FK → CLAIM	Claim generating this contribution
SOURCE_ACCUMULATOR_ID	BIGINT	FK → ACCUMULATOR	Accumulator directly receiving the contribution

TARGET_ACCUMULATOR_ID	BIGINT	FK → ACCUMULATOR	Cross-linked accumulator (e.g., family bucket)
Attributes			
CONTRIBUTION_AMOUNT	DECIMAL(10,2)	NOT NULL	Dollar amount contributed
CONTRIBUTION_DATE	DATE	NOT NULL	Date the contribution was applied
IS_CROSS_ACCUMULATOR	BOOLEAN	NOT NULL	TRUE if this row represents a cross-accumulator linkage

Notes: When an individual claim contributes to both the individual and family deductible, two rows are written: one with IS_CROSS_ACCUMULATOR = FALSE (individual bucket) and one with IS_CROSS_ACCUMULATOR = TRUE (family bucket).

3.4 Claims Domain

CLAIM

The central fact table of the database. Each row represents one adjudicated version of a pharmaceutical claim. To retrieve the final state of any claim, filter to the row where IS_CURRENT_VERSION = TRUE. Original and adjusted/reversed versions are preserved for audit purposes.

CLAIM			
Column Name	Data Type	Constraint	Description
Primary Key			
CLAIM_ID	BIGINT	PK, NOT NULL	System-generated claim row identifier (per version)
Foreign Keys			
MEMBER_SK	BIGINT	FK → MEMBER_DIM	Member version active on fill date
MEMBER_ID	VARCHAR(20)	NOT NULL, INDEX	Constant member ID
NDC	CHAR(11)	FK → DRUG_REFERENCE	11-digit NDC of dispensed drug
PRESCRIBER_ID	INT	FK → PRESCRIBER	Prescriber NPI reference
PHARMACY_ID	INT	FK → PHARMACY	Pharmacy where claim was filled
FORMULARY_DRUG_ID	BIGINT	FK → FORMULARY_DRUG	Formulary assignment at time of fill
PRIOR_AUTH_ID	BIGINT	FK → PRIOR_AUTH	Prior authorization if applicable (NULL if none)
Attributes			
CLAIM_NUMBER	VARCHAR(30)	NOT NULL, INDEX	Source system claim number (constant across adjustments)
FILL_DATE	DATE	NOT NULL	Date prescription was dispensed
DAYS_SUPPLY	INT	NOT NULL	Days supply dispensed
DAYS_SUPPLY_END_DATE	DATE	NOT NULL	Computed: FILL_DATE + DAYS_SUPPLY - 1
QUANTITY_DISPENSED	DECIMAL(10,3)	NOT NULL	Quantity actually dispensed
QUANTITY_PRESCRIBED	DECIMAL(10,3)		Quantity as written on prescription
REFILL_NUMBER	INT	NOT NULL	0 = new prescription; 1+ = refill number
IS_NEW_PRESCRIPTION	BOOLEAN	NOT NULL	TRUE if refill_number = 0
CLAIM_STATUS	VARCHAR(20)	NOT NULL	PAID, REVERSED, ADJUSTED
ADJUSTMENT_SEQ	INT	NOT NULL	0 = original; increments with each adjustment
IS_CURRENT_VERSION	BOOLEAN	NOT NULL	TRUE = final adjudicated version of this claim

REVERSAL_FLAG	BOOLEAN	NOT NULL	TRUE if this version is a reversal
ORIGINAL_CLAIM_ID	BIGINT		FK → CLAIM.CLAIM_ID of the original row (NULL if original)
SUBMITTED_DATE	DATE		Date claim was submitted for adjudication
PROCESSED_DATE	DATE		Date claim was adjudicated

Notes: CLAIM_NUMBER groups all versions of a claim. Use CLAIM_NUMBER + IS_CURRENT_VERSION = TRUE to get the final state. ORIGINAL_CLAIM_ID creates a self-referencing chain for adjustment history.

CLAIM_COST

Cost decomposition for each claim version. Captures how total claim cost is split across plan, member, and other payers.

CLAIM_COST			
Column Name	Data Type	Constraint	Description
Primary Key			
CLAIM_COST_ID	BIGINT	PK, NOT NULL	System-generated cost row identifier
Foreign Keys			
CLAIM_ID	BIGINT	FK → CLAIM, UNIQUE	One cost row per claim version
Attributes			
INGREDIENT_COST	DECIMAL(10,2)	NOT NULL	Drug ingredient cost (AWP-based)
DISPENSING_FEE	DECIMAL(10,2)		Pharmacy dispensing fee
SALES_TAX	DECIMAL(10,2)		Applicable sales tax
GROSS_AMOUNT_DUE	DECIMAL(10,2)	NOT NULL	Total amount due before cost sharing
PLAN_PAID_AMOUNT	DECIMAL(10,2)	NOT NULL	Amount paid by the health plan
MEMBER_COPAY	DECIMAL(10,2)	NOT NULL	Member fixed copay amount
MEMBER_COINSURANCE	DECIMAL(10,2)		Member coinsurance amount
MEMBER_DEDUCTIBLE	DECIMAL(10,2)		Amount applied to member deductible
MEMBER_TOTAL_PAID	DECIMAL(10,2)	NOT NULL	Total member out-of-pocket for this claim
OTHER_PAYER_AMOUNT	DECIMAL(10,2)		Amount paid by other payer (COB)
OTHER_PAYER_ID	VARCHAR(30)		Identifier of other payer if applicable
BASIS_OF_REIMBURSEMENT	VARCHAR(20)		AWP, MAC, U&C, Negotiated, etc.

Notes: PLAN_PAID + MEMBER_TOTAL_PAID + OTHER_PAYER = GROSS_AMOUNT_DUE. Each field should net to zero on a reversal row.

CLAIM_ADJUSTMENT

Audit log of adjustment and reversal events against a claim. Provides a human-readable reason and timestamp for each version transition.

CLAIM_ADJUSTMENT			
Column Name	Data Type	Constraint	Description
Primary Key			
ADJUSTMENT_ID	BIGINT	PK, NOT NULL	System-generated adjustment event identifier
Foreign Keys			
ORIGINAL_CLAIM_ID	BIGINT	FK → CLAIM	The original claim row this event relates to
NEW_CLAIM_ID	BIGINT	FK → CLAIM	The new claim version created by this event
Attributes			
ADJUSTMENT_TYPE	VARCHAR(20)	NOT NULL	REVERSAL, CORRECTION, REPROCESS
ADJUSTMENT_REASON	VARCHAR(200)		Free-text or coded reason for adjustment
ADJUSTMENT_DATE	DATE	NOT NULL	Date the adjustment was processed
PROCESSED_BY	VARCHAR(50)		User or system that processed the adjustment
CREATED_AT	TIMESTAMP	NOT NULL	Row creation timestamp

3.5 Drug Reference Domain

DRUG_REFERENCE

Internal drug master table mapping each 11-digit NDC to its full GPI-14 and descriptive attributes. Package and unit information supports days supply validation and adherence calculations.

DRUG_REFERENCE			
Column Name	Data Type	Constraint	Description
Primary Key			
NDC	CHAR(11)	PK, NOT NULL	11-digit National Drug Code (no dashes)
Attributes			
GPI_14	CHAR(14)	NOT NULL, INDEX	Full 14-digit Generic Product Identifier
GPI_2	CHAR(2)	NOT NULL	Drug group level
GPI_4	CHAR(4)	NOT NULL	Drug class level
GPI_6	CHAR(6)	NOT NULL	Drug name level
GPI_8	CHAR(8)	NOT NULL	Drug name extended level
GPI_10	CHAR(10)	NOT NULL	Dosage form level
DRUG_NAME	VARCHAR(100)	NOT NULL	Full drug product name
GENERIC_NAME	VARCHAR(100)		Generic (INN) name
BRAND_NAME	VARCHAR(100)		Brand name if applicable
MANUFACTURER	VARCHAR(100)		Drug manufacturer / labeler
STRENGTH	VARCHAR(50)		Drug strength (e.g., 10 mg)
DOSAGE_FORM	VARCHAR(50)		Tablet, Capsule, Solution, etc.
ROUTE_OF_ADMINISTRATION	VARCHAR(50)		Oral, Topical, Injectable, etc.
PACKAGE_SIZE	DECIMAL(10,3)		Units per package
PACKAGE_SIZE_UOM	VARCHAR(20)		Unit of measure for package size
UNIT_DOSE_SIZE	DECIMAL(10,5)		Amount per unit dose
UNIT_DOSE_UOM	VARCHAR(20)		Unit of measure for unit dose
DEA_SCHEDULE	CHAR(2)		DEA controlled substance schedule if applicable
GENERIC_FLAG	BOOLEAN	NOT NULL	TRUE if generic product
SPECIALTY_FLAG	BOOLEAN	NOT NULL	TRUE if specialty drug
EFFECTIVE_DATE	DATE	NOT NULL	Date this NDC became active
EXPIRATION_DATE	DATE		Date this NDC was discontinued (NULL = active)
IS_ACTIVE	BOOLEAN	NOT NULL	TRUE if NDC is currently marketed

Notes: GPI hierarchy fields are derived from GPI_14 but stored explicitly for query performance. Index on GPI_14 and GPI_10 to support therapeutic grouping joins.

PDC_THERAPEUTIC_GROUP

Defines therapeutic cohorts used for PDC calculation. NDCs are grouped into cohorts so that drug switches within a class are measured together for adherence purposes.

PDC_THERAPEUTIC_GROUP			
Column Name	Data Type	Constraint	Description
Primary Key			
GROUP_ID	INT	PK, NOT NULL	System-generated group identifier
Attributes			
GROUP_CODE	VARCHAR(30)	NOT NULL, UNIQUE	Short code for the therapeutic group
GROUP_NAME	VARCHAR(100)	NOT NULL	Descriptive name (e.g., Statins, ACE Inhibitors)
THERAPY_CLASS	VARCHAR(100)		Broader therapy class description
GPI_4_BASE	CHAR(4)		GPI-4 that anchors this group (for reference)
STAR_MEASURE_FLAG	BOOLEAN		TRUE if this group maps to a CMS Star Measures metric
DESCRIPTION	VARCHAR(500)		Clinical description of the group
CREATED_AT	TIMESTAMP	NOT NULL	Row creation timestamp

PDC_NDC_GROUP_MAP

Maps individual NDCs to therapeutic groups. An NDC may belong to more than one group if clinically appropriate. Effective dates support formulary and drug changes over time.

PDC_NDC_GROUP_MAP			
Column Name	Data Type	Constraint	Description
Primary Key			
MAP_ID	BIGINT	PK, NOT NULL	System-generated mapping identifier
Foreign Keys			
NDC	CHAR(11)	FK → DRUG_REFERENCE	NDC being mapped
GROUP_ID	INT	FK → PDC_THERAPEUTIC_GROUP	Target therapeutic group
Attributes			
EFFECTIVE_DATE	DATE	NOT NULL	Date this mapping became active
EXPIRATION_DATE	DATE		Date this mapping expired (NULL = active)
IS_ACTIVE	BOOLEAN	NOT NULL	TRUE if mapping is currently active

CREATED_AT	TIMESTAMP	NOT NULL	Row creation timestamp
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Notes: Composite unique constraint on (NDC, GROUP_ID, EFFECTIVE_DATE) to prevent duplicate active mappings.

3.6 Provider Domain

PRESCRIBER

PRESCRIBER			
Column Name	Data Type	Constraint	Description
Primary Key			
PRESCRIBER_ID	INT	PK, NOT NULL	System-generated prescriber identifier
Attributes			
NPI	CHAR(10)	NOT NULL, UNIQUE	National Provider Identifier
FIRST_NAME	VARCHAR(50)	NOT NULL	Prescriber first name
LAST_NAME	VARCHAR(50)	NOT NULL	Prescriber last name
CREDENTIAL	VARCHAR(30)		MD, DO, NP, PA, etc.
PRIMARY_SPECIALTY	VARCHAR(100)		Primary specialty description
SECONDARY_SPECIALTY	VARCHAR(100)		Secondary specialty if applicable
TAXONOMY_CODE	VARCHAR(20)		NUCC taxonomy code
DEA_NUMBER	VARCHAR(15)		DEA number for controlled substances
ADDRESS_LINE1	VARCHAR(100)		Practice street address
ADDRESS_LINE2	VARCHAR(100)		Suite / unit
CITY	VARCHAR(50)		City
STATE	CHAR(2)		State code
ZIP_CODE	VARCHAR(10)		ZIP code
PHONE	VARCHAR(15)		Practice phone number
ACTIVE_FLAG	BOOLEAN	NOT NULL	TRUE if prescriber is active
CREATED_AT	TIMESTAMP	NOT NULL	Row creation timestamp
UPDATED_AT	TIMESTAMP	NOT NULL	Last update timestamp

PHARMACY

PHARMACY			
Column Name	Data Type	Constraint	Description
Primary Key			
PHARMACY_ID	INT	PK, NOT NULL	System-generated pharmacy identifier
Attributes			
NPI	CHAR(10)	NOT NULL, UNIQUE	National Provider Identifier
PHARMACY_NAME	VARCHAR(100)	NOT NULL	Pharmacy name
PHARMACY_TYPE	VARCHAR(30)	NOT NULL	RETAIL, MAIL_ORDER, SPECIALTY, COMPOUNDING, LONG_TERM_CARE

CHAIN_CODE	VARCHAR(20)		Chain identifier if part of a pharmacy chain
NABP_NUMBER	VARCHAR(15)		NABP dispensing site number
ADDRESS_LINE1	VARCHAR(100)		Street address
CITY	VARCHAR(50)		City
STATE	CHAR(2)		State code
ZIP_CODE	VARCHAR(10)		ZIP code
PHONE	VARCHAR(15)		Pharmacy phone
ACTIVE_FLAG	BOOLEAN	NOT NULL	TRUE if pharmacy is currently in network
CREATED_AT	TIMESTAMP	NOT NULL	Row creation timestamp
UPDATED_AT	TIMESTAMP	NOT NULL	Last update timestamp

Notes: PHARMACY_TYPE drives days supply logic: MAIL_ORDER and SPECIALTY often allow 90-day supplies which affect PDC calculations.

3.7 Authorization Domain

PRIOR_AUTH

Tracks the full lifecycle of prior authorization and formulary exception requests. Claims that are approved under a PA or exception reference this table to explain cost sharing treatment that deviates from the standard formulary.

PRIOR_AUTH			
Column Name	Data Type	Constraint	Description
Primary Key			
PRIOR_AUTH_ID	BIGINT	PK, NOT NULL	System-generated PA identifier
Foreign Keys			
MEMBER_ID	VARCHAR(20)	NOT NULL, INDEX	Member the PA applies to
NDC	CHAR(11)	FK → DRUG_REFERENCE	Drug approved under this PA
PRESCRIBER_ID	INT	FK → PRESCRIBER	Requesting prescriber
FORMULARY_DRUG_ID	BIGINT	FK → FORMULARY_DRUG	Formulary drug record this overrides
Attributes			
AUTH_TYPE	VARCHAR(30)	NOT NULL	PRIOR_AUTH, STEP_THERAPY_OVERRIDE, FORMULARY_EXCEPTION, QTY_LIMIT_OVERRIDE
REQUEST_DATE	DATE	NOT NULL	Date the authorization was requested
STATUS	VARCHAR(20)	NOT NULL	PENDING, APPROVED, DENIED, APPEALED, EXPIRED
DECISION_DATE	DATE		Date the authorization decision was made
APPROVING_CLINICIAN	VARCHAR(100)		Name or ID of clinician who approved
DENIAL_REASON	VARCHAR(200)		Coded or free-text denial reason
EFFECTIVE_DATE	DATE		Date the authorization becomes effective
EXPIRATION_DATE	DATE		Date the authorization expires
APPROVED_QUANTITY	DECIMAL(10,3)		Quantity approved per fill if quantity-limited PA
APPROVED_DAYS_SUPPLY	INT		Days supply approved per fill
APPEAL_DATE	DATE		Date of appeal if status = APPEALED
APPEAL_OUTCOME	VARCHAR(20)		UPHELD, OVERTURNED
NOTES	VARCHAR(1000)		Clinical or administrative notes
CREATED_AT	TIMESTAMP	NOT NULL	Row creation timestamp
UPDATED_AT	TIMESTAMP	NOT NULL	Last update timestamp

Notes: An approved PRIOR_AUTH is linked from CLAIM.PRIOR_AUTH_ID when the claim is processed under an exception. Multiple claims may reference the same PA within its effective period.

4. Entity Relationships

From Entity	Cardinality	To Entity	Description
MEMBER_DIM	1 : Many	MEMBER_ENROLLMENT	A member version may have multiple enrollment periods
MEMBER_DIM	1 : Many	CLAIM	A member version is referenced by many claims (via MEMBER_SK)
MEMBER_DIM	1 : Many (via MEMBER_ID)	ACCUMULATOR	A member has accumulator buckets per benefit period
EMPLOYER	1 : Many	MEMBER_DIM	An employer is associated with many member versions
HEALTH_PLAN	1 : Many	MEMBER_ENROLLMENT	A plan has many member enrollments
HEALTH_PLAN	1 : Many	FORMULARY	A plan has multiple versioned formularies over time
FORMULARY	1 : Many	FORMULARY_DRUG	A formulary version has many drug-tier assignments
FORMULARY	1 : Many	FORMULARY_TIER	A formulary version has many tier cost-sharing rules
FORMULARY_TIER	1 : Many	FORMULARY_DRUG	A tier has many drugs assigned to it
DRUG_REFERENCE	1 : Many	CLAIM	An NDC is referenced by many claims
DRUG_REFERENCE	1 : Many	FORMULARY_DRUG	An NDC appears in many formulary versions
DRUG_REFERENCE	1 : Many	PDC_NDC_GROUP_MAP	An NDC maps to one or more therapeutic groups
PDC_THERAPEUTIC_GROUP	1 : Many	PDC_NDC_GROUP_MAP	A group contains many NDC mappings
PRESCRIBER	1 : Many	CLAIM	A prescriber appears on many claims
PRESCRIBER	1 : Many	PRIOR_AUTH	A prescriber submits many PA requests
PHARMACY	1 : Many	CLAIM	A pharmacy fills many claims
CLAIM	1 : 1	CLAIM_COST	Each claim version has exactly one cost decomposition row
CLAIM	1 : Many	CLAIM_ADJUSTMENT	An original claim may have many adjustment events
CLAIM	1 : Many	ACCUMULATOR_CONTRIBUTION	A claim generates one or more accumulator contributions
ACCUMULATOR	1 : Many	ACCUMULATOR_CONTRIBUTION	An accumulator receives many contributions over time
PRIOR_AUTH	1 : Many	CLAIM	An approved PA may be referenced by multiple claims within its effective period
MEMBER_ENROLLMENT	1 : Many	ACCUMULATOR	An enrollment period has individual and family accumulator buckets

5. Key Analytics Patterns

5.1 Retrieving Final Adjudicated Claims

Always filter CLAIM on IS_CURRENT_VERSION = TRUE to exclude reversed and superseded versions. Join CLAIM_COST to decompose costs. This pattern should be applied as a base view or CTE in all analytical queries.

5.2 PDC Calculation

To calculate PDC for a therapeutic group: (1) join CLAIM to PDC_NDC_GROUP_MAP on NDC and GROUP_ID for the target cohort, filtering to active mappings at fill date; (2) filter to IS_CURRENT_VERSION = TRUE; (3) for each member, build a day-level coverage calendar using FILL_DATE and DAYS_SUPPLY_END_DATE; (4) handle overlapping fills by capping to the measurement period end date; (5) $PDC = \text{unique covered days} / \text{total days in measurement period}$.

5.3 Point-in-Time Member Version Lookup

To find the member version active at a given date: join MEMBER_DIM on MEMBER_ID where EFFECTIVE_DATE <= target_date AND (EXPIRATION_DATE IS NULL OR EXPIRATION_DATE >= target_date). Alternatively, join CLAIM on MEMBER_SK directly, which already resolves to the correct version at fill date.

5.4 Formulary Compliance at Fill Date

To determine a claim's formulary status at fill date: (1) resolve the member's health plan from MEMBER_ENROLLMENT at fill date; (2) find the active FORMULARY version for that plan at fill date; (3) look up the FORMULARY_DRUG row for the claim NDC within that formulary version where EFFECTIVE_DATE <= fill_date AND (EXPIRATION_DATE IS NULL OR EXPIRATION_DATE >= fill_date); (4) join FORMULARY_TIER to get cost sharing rules. If a PRIOR_AUTH_ID exists on the claim, the PA record overrides standard formulary treatment.

5.5 Accumulator Balance at Point in Time

Sum ACCUMULATOR_CONTRIBUTION.CONTRIBUTION_AMOUNT where IS_CROSS_ACCUMULATOR matches the bucket type (FALSE for individual direct contributions, TRUE for family cross-contributions), grouped by ACCUMULATOR_ID and filtered to CONTRIBUTION_DATE <= target_date. Compare to ACCUMULATOR.LIMIT_AMOUNT to determine remaining deductible or OOP balance.

6. Implementation Notes

6.1 Indexing Recommendations

- CLAIM: index on (MEMBER_ID, FILL_DATE), (CLAIM_NUMBER), (NDC), (IS_CURRENT_VERSION)
- MEMBER_DIM: index on (MEMBER_ID), (EFFECTIVE_DATE, EXPIRATION_DATE)
- DRUG_REFERENCE: index on (GPI_14), (GPI_4)
- PDC_NDC_GROUP_MAP: index on (NDC, GROUP_ID, EFFECTIVE_DATE)
- FORMULARY_DRUG: index on (FORMULARY_ID, NDC)
- ACCUMULATOR_CONTRIBUTION: index on (ACCUMULATOR_ID, CONTRIBUTION_DATE)

6.2 SCD2 Maintenance

When a member attribute changes: (1) UPDATE the current row — set IS_CURRENT = FALSE, EXPIRATION_DATE = new_effective_date - 1 day; (2) INSERT a new row with the updated attributes, IS_CURRENT = TRUE, EXPIRATION_DATE = NULL, EFFECTIVE_DATE = change date. New claims should resolve MEMBER_SK at load time by matching MEMBER_ID and fill date against the SCD2 effective/expiration range.

6.3 Claim Adjustment Processing

When a claim is adjusted or reversed: (1) UPDATE the original CLAIM row — set IS_CURRENT_VERSION = FALSE; (2) INSERT a new CLAIM row with the corrected data, ADJUSTMENT_SEQ incremented, IS_CURRENT_VERSION = TRUE, ORIGINAL_CLAIM_ID pointing to the original; (3) INSERT a CLAIM_ADJUSTMENT row linking original to new with reason and date; (4) INSERT or reverse CLAIM_COST and ACCUMULATOR_CONTRIBUTION rows accordingly.

6.4 NDC Format

Store all NDCs as 11-digit zero-padded strings without dashes (e.g., 00069315041). Standardize at load time. Source systems may deliver NDCs in 5-4-2, 5-3-2, or 4-4-2 formats — apply appropriate zero-padding to each segment before storage.